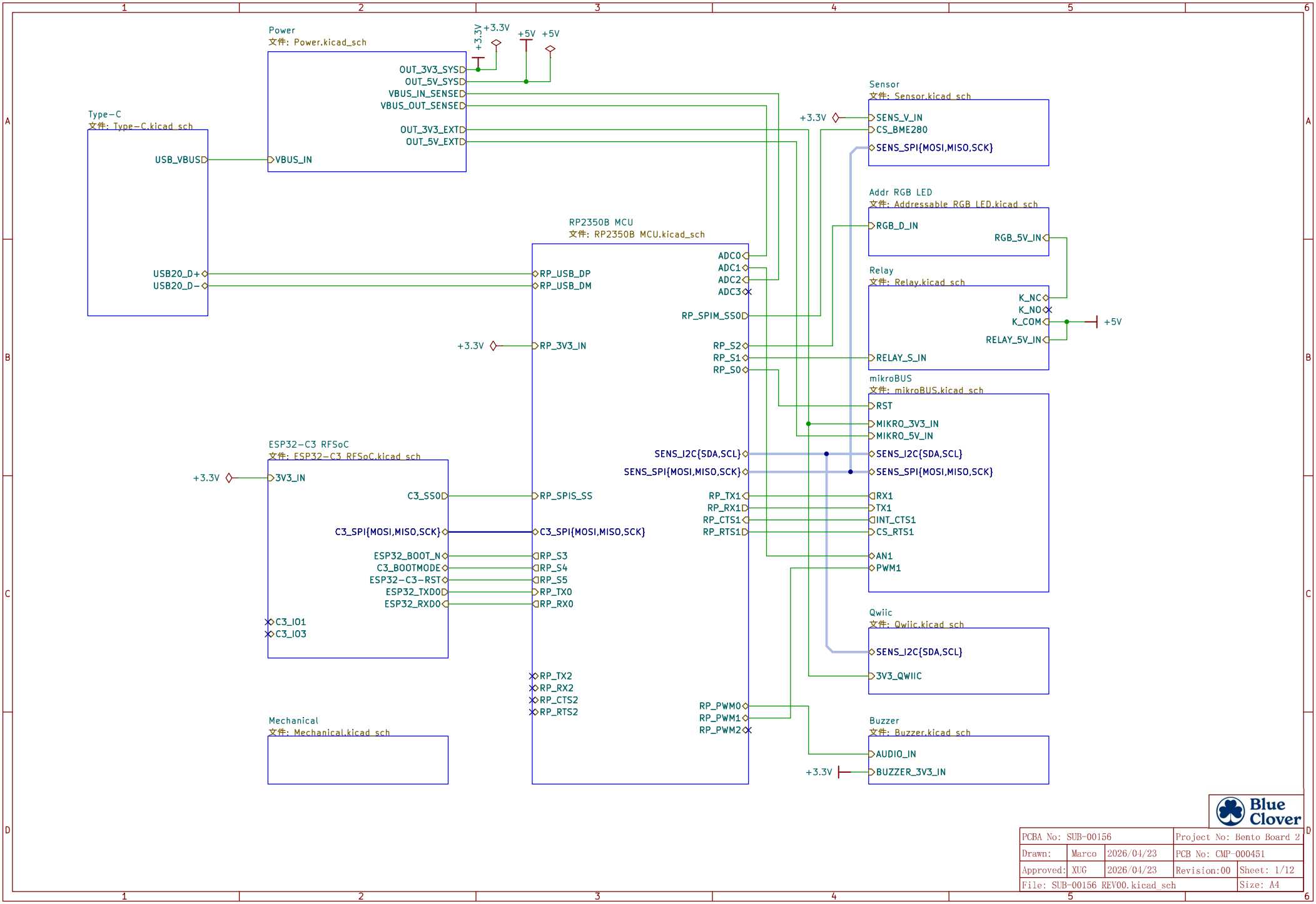
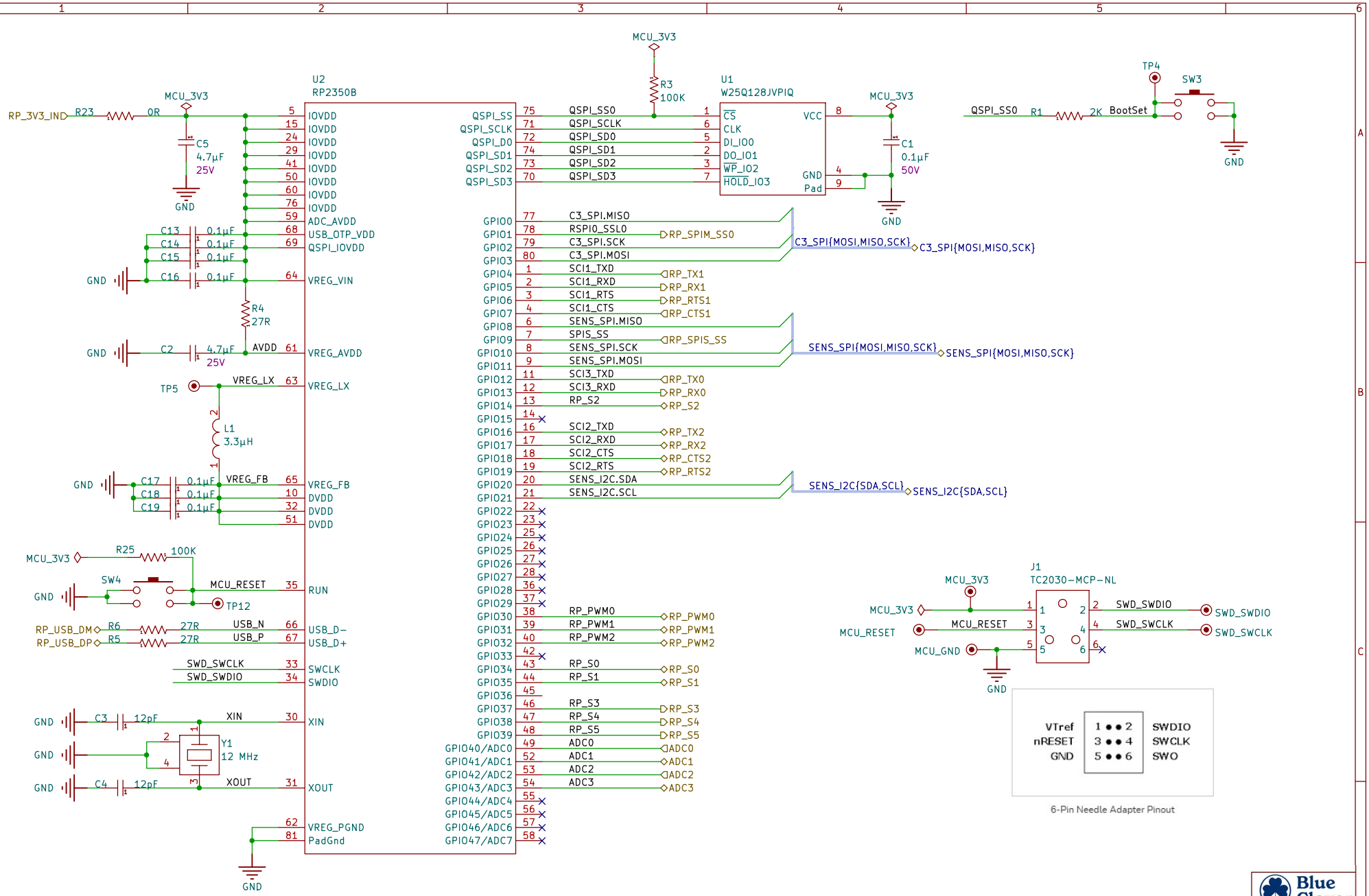
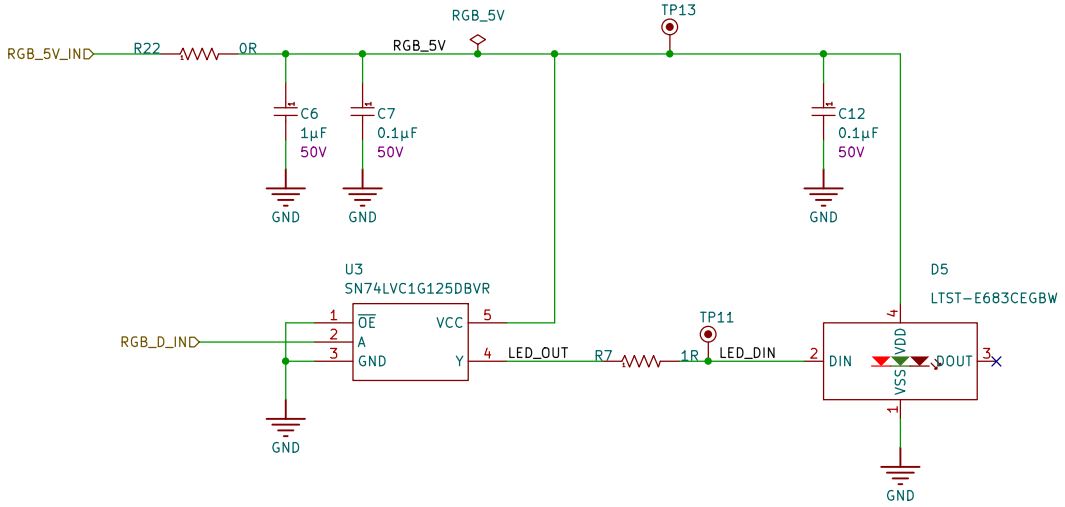


Blue Clover

PCBA No: SUB-00156			Project No: Bento Board 2	
Drawn:	Marco	2026/04/23	PCB No: CMP-000451	
Approved:	XUG	2026/04/23	Revision:00	Sheet: 1/12
File: SUB-00156 REV00.kicad_sch				Size: A4

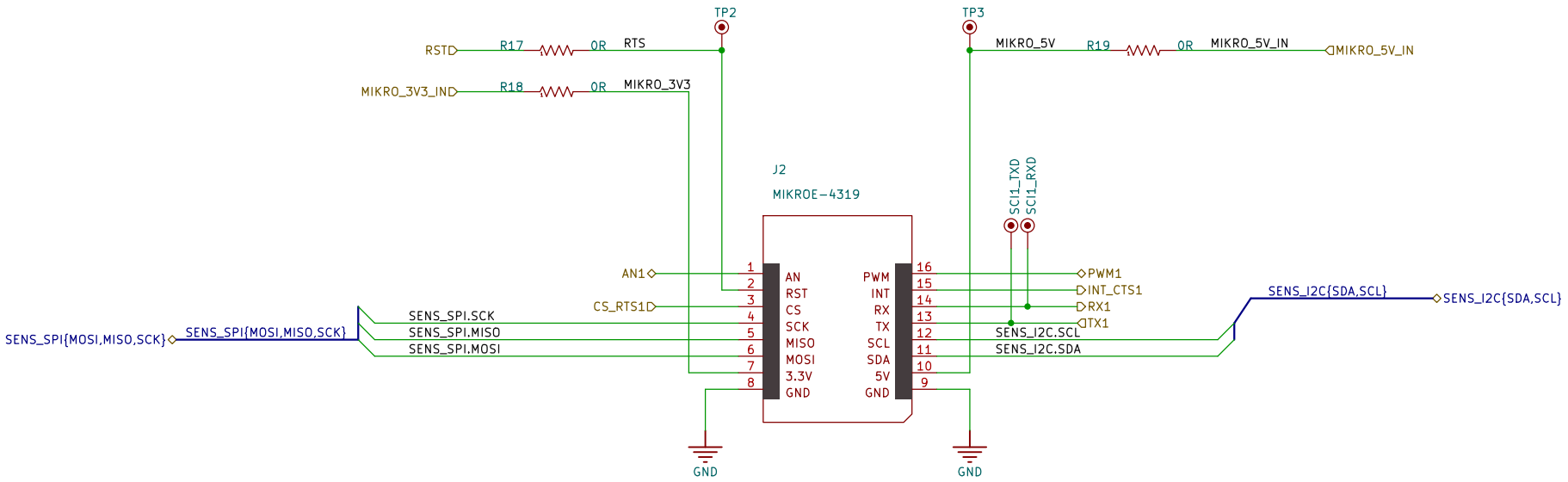


PCBA No: SUB-00156			Project No: Bento Board 2	
Drawn:	Marco	2026/04/23	PCB No: CMP-000451	
Approved:	XUG	2026/04/23	Revision:00	Sheet: 2/12
File: RP2350B MCU.kicad_sch				Size: A4



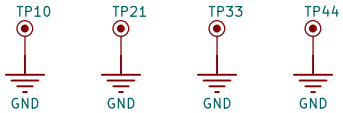
PCBA No: SUB-00156			Project No: Bento Board 2	
Drawn:	Marco	2026/04/23	PCB No: CMP-000451	
Approved:	XUG	2026/04/23	Revision:00	Sheet: 3/12
File: Addressable RGB LED.kicad_sch				Size: A4

MikroE Click mikroBUS□



Blue Clover

PCBA No: SUB-00156			Project No: Bento Board 2	
Drawn:	Marco	2026/04/23	PCB No: CMP-000451	
Approved:	XUG	2026/04/23	Revision:00	Sheet: 4/12
File: mikroBUS.kicad_sch				Size: A4







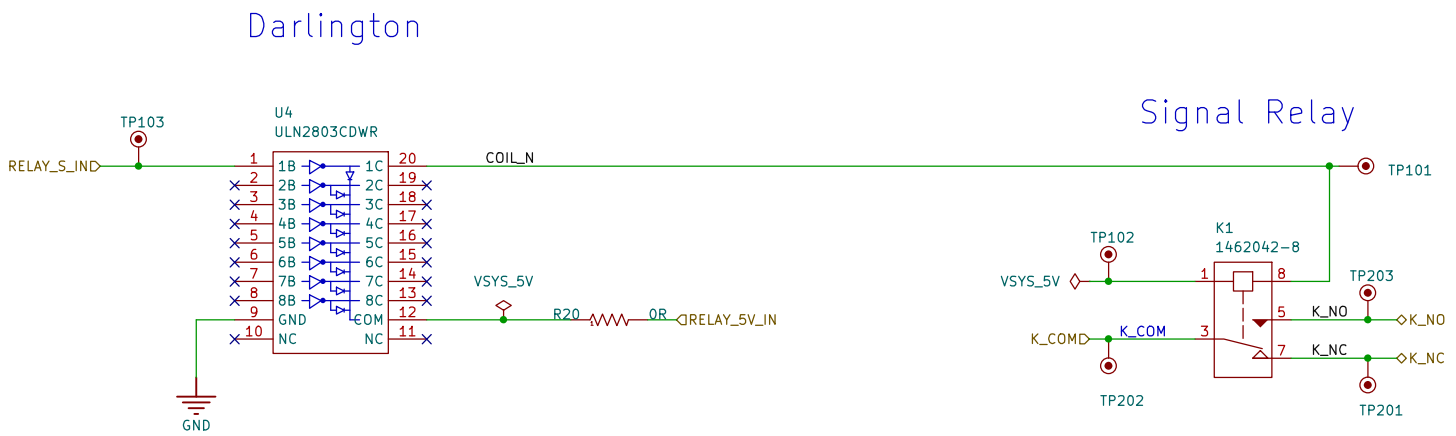
Blank label
8*12mm

Blank label
12*24mm



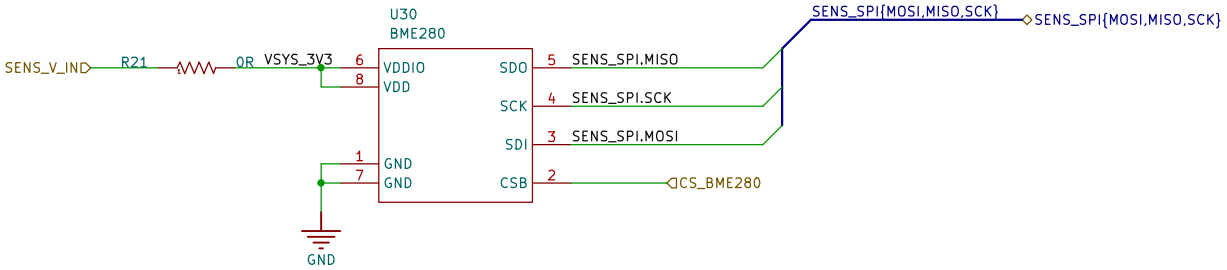
CMP-000451 REV:00

PCBA No: SUB-00156			Project No: Bento Board 2	
Drawn:	Marco	2026/04/23	PCB No: CMP-000451	
Approved:	XUG	2026/04/23	Revision:00	Sheet: 5/12
File: Mechanical.kicad_sch				Size: A4

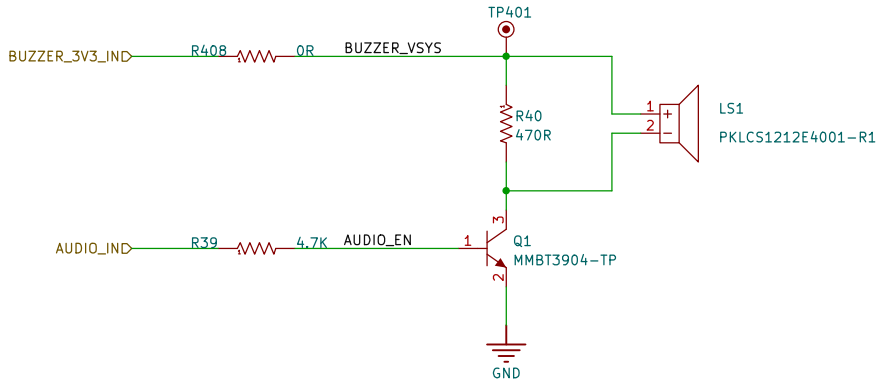


PCBA No: SUB-00156			Project No: Bento Board 2	
Drawn:	Marco	2026/04/23	PCB No: CMP-000451	
Approved:	XUG	2026/04/23	Revision:00	Sheet: 6/12
File: Relay.kicad_sch				Size: A4

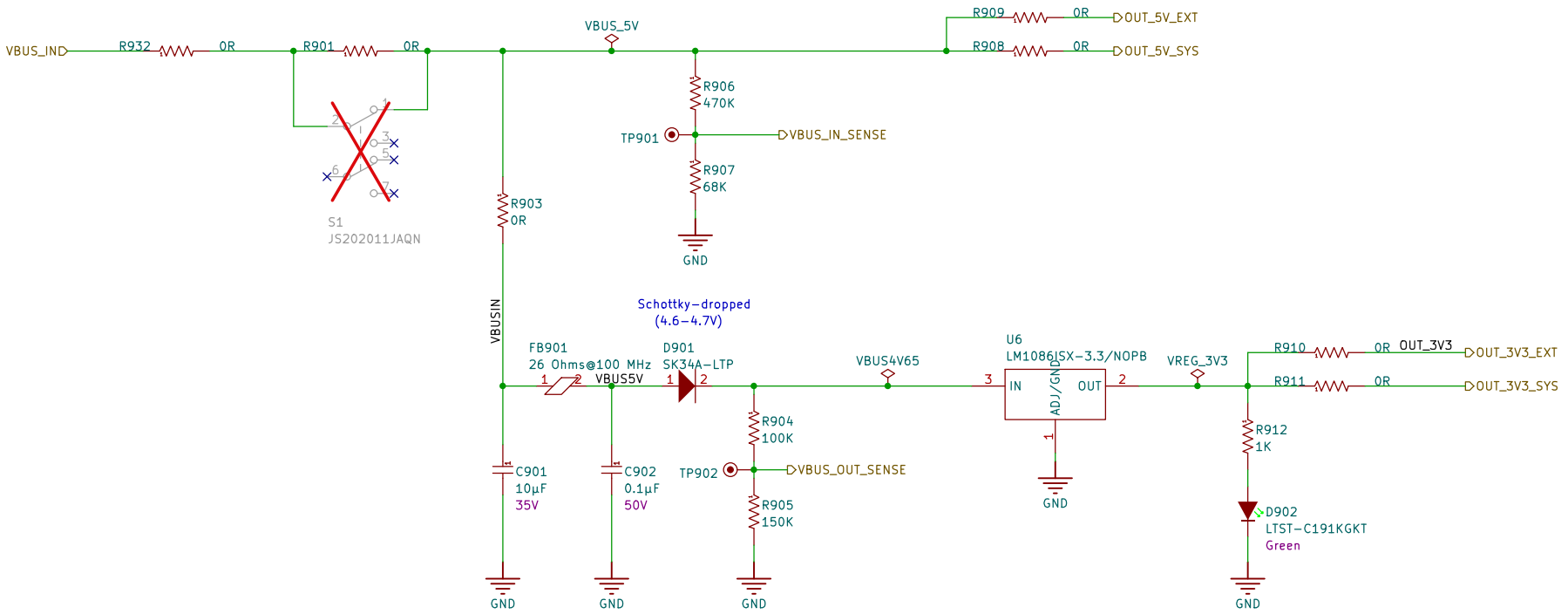
Temperature/Humidity/Pressure



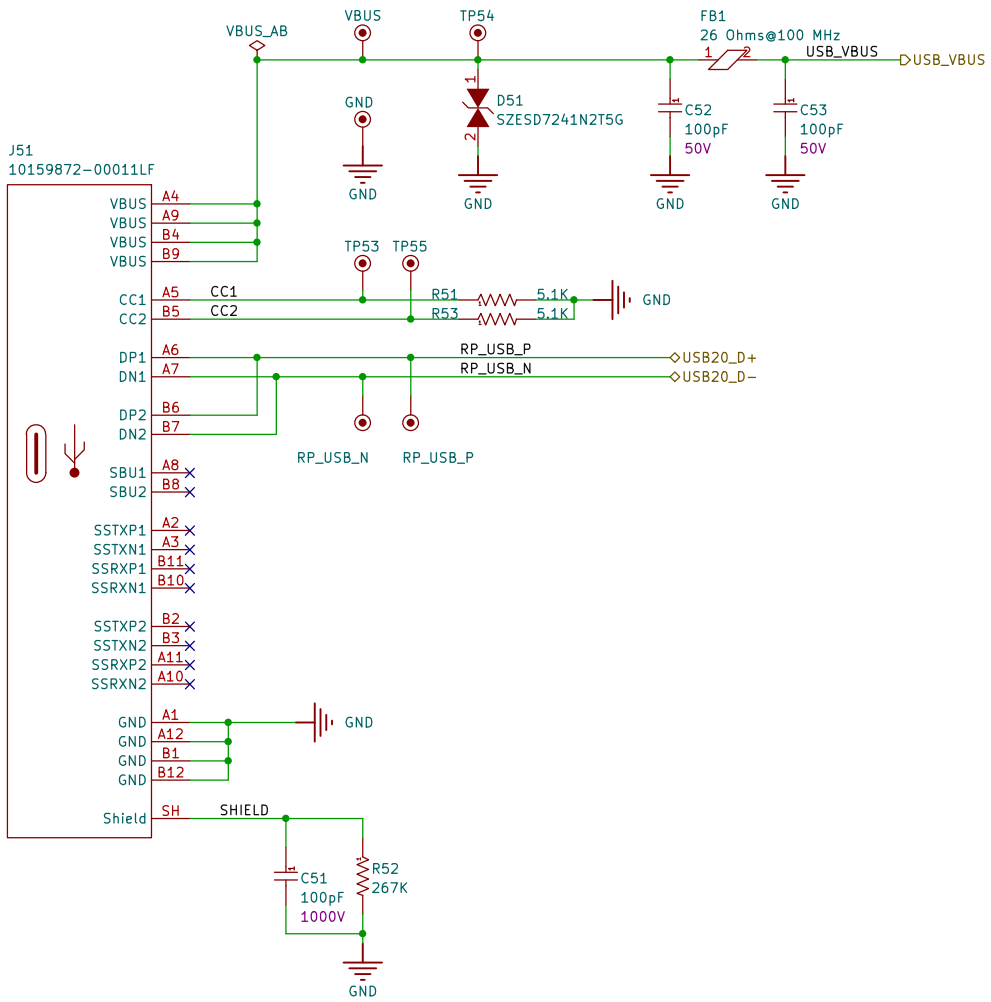
PCBA No: SUB-00156			Project No: Bento Board 2	
Drawn:	Marco	2026/04/23	PCB No: CMP-000451	
Approved:	XUG	2026/04/23	Revision:00	Sheet: 7/12
File: Sensor.kicad_sch				Size: A4



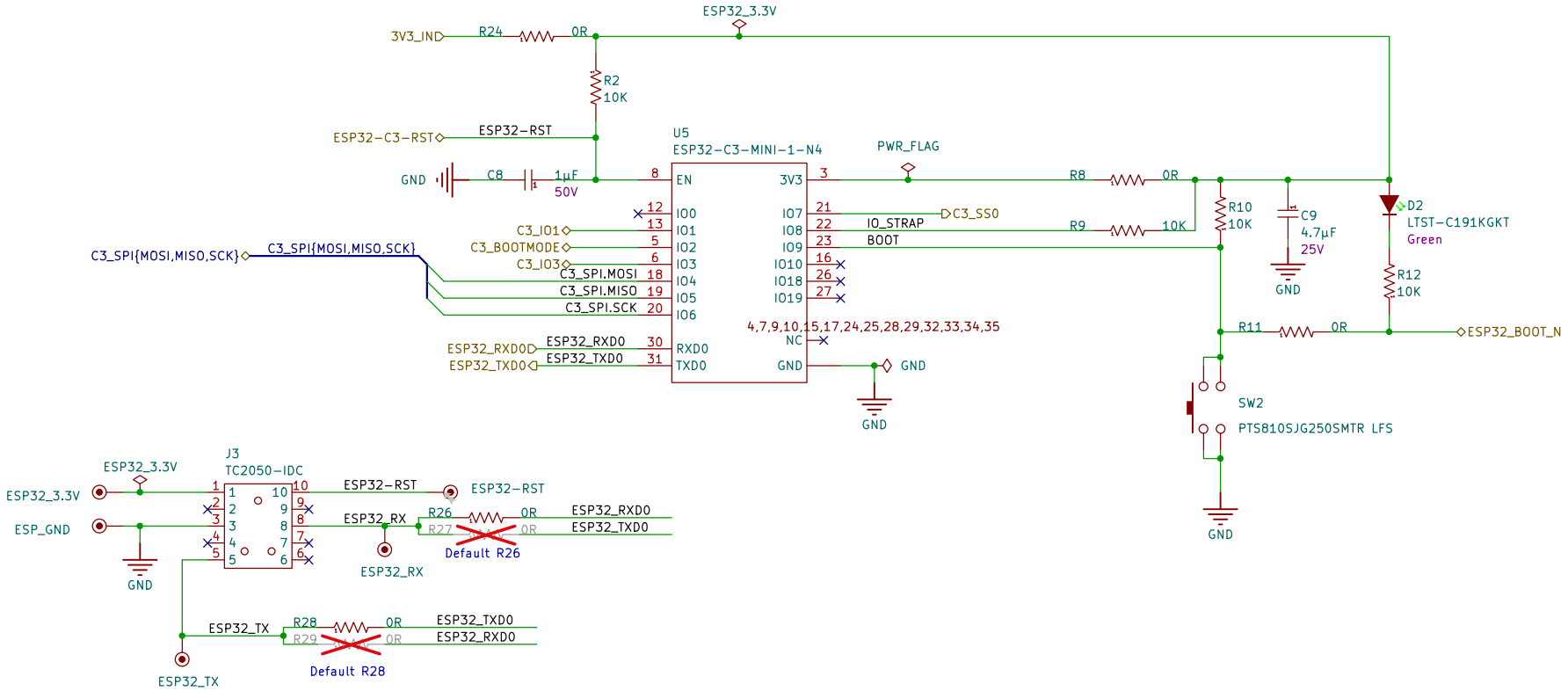
PCBA No: SUB-00156			Project No: Bento Board 2	
Drawn:	Marco	2026/04/23	PCB No: CMP-000451	
Approved:	XUG	2026/04/23	Revision:00	Sheet: 8/12
File: Buzzer.kicad_sch				Size: A4



PCBA No: SUB-00156			Project No: Bento Board 2	
Drawn:	Marco	2026/04/23	PCB No: CMP-000451	
Approved:	XUG	2026/04/23	Revision:00	Sheet: 9/12
File: Power.kicad_sch				Size: A4



PCBA No: SUB-00156			Project No: Bento Board 2	
Drawn:	Marco	2026/04/23	PCB No: CMP-000451	
Approved:	XUG	2026/04/23	Revision:00	Sheet: 10/12
File: Type-C.kicad_sch				Size: A4



Bluetooth LE, WiFi 2.4GHz

The J3 (TC2050-IDC) interface definition
is not the SWD standard connection method.

https://kb.segger.com/20-pin_J-Link_Connector#Pinout_for_SWD_+_Virtual_COM_Port_(VCOM)

https://www.segger.com/supported-devices/espressif/esp32-c6/esp32-c6/

Program Interconnect

ESP32-C3 Pin	Direction	ESP32-C Function
GPIO20	RP → ESP32	RXD0
GPIO21	ESP32 → RP	TXD0
EN	RP → ESP32	Reset
GPIO2	RP → ESP32	Boot mode(IO2)

ESP32-C3 GPIO mapping

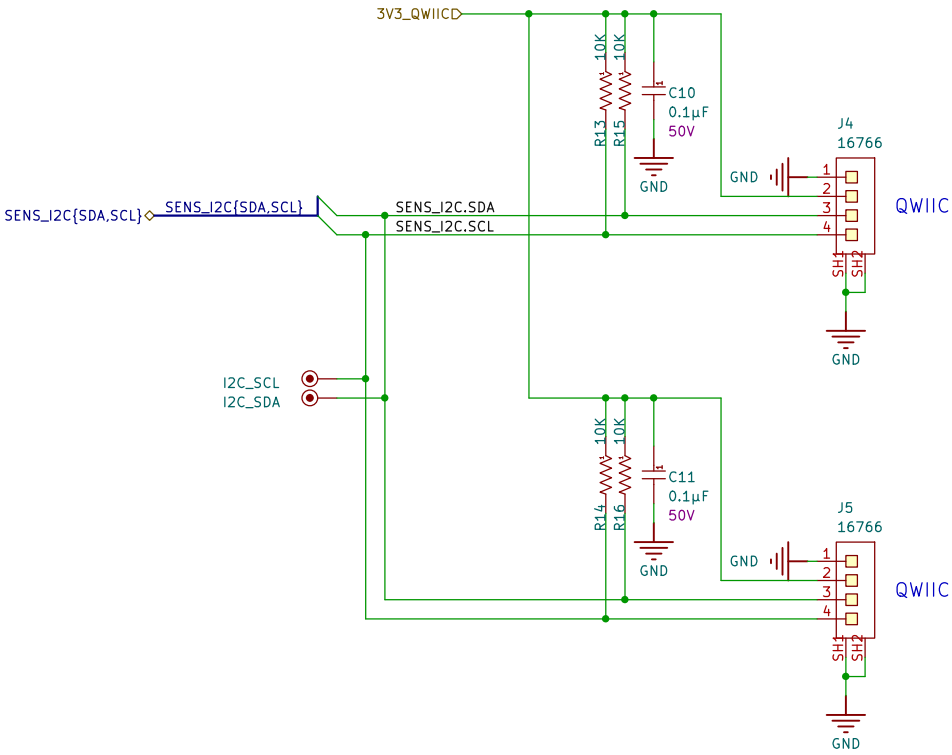
IO0	NC	Strapping pin
IO1	GPIO1	General Purpose
IO2	MOOTMODE	Pull High to enter UART download
IO3	GPIO3	General Purpose
IO4	MOSI	SPI
IO5	MISO	SPI
IO6	SCLK	SPI
IO7	CS0	SPI
IO8	IO_STRAP	Strapping pin
IO9	BOOT	Pull low to enter bootloader
IO10	FLASH_STRAP	Flash voltage strapping pin
IO18	GPIO18	General Purpose
IO19	GPIO19	General Purpose
IO20	UART0_RXD	Serial input
IO21	UART0_TXD	Serial output

Blue Clover

PCBA No: SUB-00156			Project No: Bento Board 2	
Drawn:	Marco	2026/04/23	PCB No: CMP-000451	
Approved:	XUG	2026/04/23	Revision:00	Sheet: 11/12
File: ESP32-C3_RFSoC.kicad_sch				Size: A4

You must use the same style 4-pin connector,JST or equivalent;
you may not use a 5-pin or a different size connector.

Boards must be 3.3V.



PCBA No: SUB-00156			Project No: Bento Board 2	
Drawn:	Marco	2026/04/23	PCB No: CMP-000451	
Approved:	XUG	2026/04/23	Revision:00	Sheet: 12/12
File: Qwiic.kicad_sch				Size: A4